A Digital Twin and Graph Neural Network-Based Framework for Fault Prediction and Optimization in 5G/6G Networks

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***Abstract* - The increasing complexity and scale of next-generation wireless networks such as 5G and 6G demand intelligent, adaptive, and autonomous management systems. Traditional network control mechanisms, which rely on rule-based monitoring and static configuration, are no longer sufficient to handle the rapidly changing topologies, high device densities, and diverse service requirements of these modern communication environments. This paper presents a MATLAB-based simulation framework that integrates a Digital Twin (DT) model with Graph Neural Networks (GNNs) for proactive fault prediction and autonomous network optimization. The Digital Twin serves as a real-time virtual mirror of the physical network, continuously updated with telemetry data such as node latency, bandwidth usage, and failure events. The GNN is trained on graph-structured network data to identify fault-prone nodes and congestion areas before performance degradation occurs. Once anomalies are detected, intelligent optimization algorithms are applied to reroute traffic and reallocate bandwidth, allowing the system to self-heal and adapt in real time. Simulation results show significant improvements in network performance, including reduced latency, increased bandwidth utilization, and high fault prediction accuracy as confirmed by confusion matrix evaluation. The proposed framework lays a scalable foundation for the development of resilient, self-optimizing, and AI-driven 5G/6G communication networks, with potential applications in autonomous base stations, mission-critical IoT, smart cities, and intelligent transportation systems.**

***Keywords*— Digital Twin (DT), Graph Neural Networks (GNNs), Fault Prediction, 5G/6G Networks, Network Optimization.**

I. INTRODUCTION

The evolution of wireless communication from 1G to 5G and beyond has driven a technological revolution that impacts virtually every aspect of modern life.

The fifth-generation (5G) networks, and the rapidly emerging sixth-generation (6G) networks, aim to fulfill the increasing demands for faster data rates, ultra-low latency, massive device connectivity, and enhanced reliability. As the density and heterogeneity of these networks expand, traditional network management systems face unprecedented challenges in maintaining service quality, operational efficiency, and fault resilience. Manual configuration, static rules, and human-in-the-loop monitoring can no longer scale to meet the real-time requirements of mission-critical applications, including autonomous vehicles, remote surgery, augmented reality, and massive IoT deployments.

Next-generation networks introduce complex topologies with millions of interconnected elements that dynamically interact with each other. These networks experience highly variable traffic conditions, hardware and software failures, and frequent topology changes due to node mobility and resource reallocation. Consequently, network operators must shift towards intelligent, adaptive, and autonomous systems capable of predicting faults and optimizing resources without human intervention. To address these needs, this paper presents a comprehensive MATLAB-based framework that integrates a Digital Twin (DT) simulation environment with Graph Neural Networks (GNNs) for autonomous fault prediction and real-time optimization in 5G/6G networks.

This framework is designed to support real-time replication of physical network conditions through the DT, while leveraging the power of GNNs to process graph-structured data and identify early signs of faults and performance degradation.

Unlike conventional monitoring systems, our approach provides a proactive layer of intelligence that anticipates failures and adapts dynamically by rerouting traffic and reallocating bandwidth. The framework is modular, scalable, and adaptable, providing a robust foundation for self-healing and self-optimizing communication infrastructures.

The subsequent sections detail the core components of this framework, including the design and functionality of the Digital Twin, the architecture of the GNN models, the strategies for fault prediction, and the MATLAB implementation that enables system simulation, visualization, and performance evaluation.

A. *Digital Twin (DT)*

A Digital Twin (DT) is a dynamic, virtual representation of a physical system that is continuously updated using real-time data. In the context of network management, a DT serves as a mirror of the actual communication network, capturing its current state, historical trends, and predictive behaviors. This virtual environment enables network administrators and automated systems to monitor, analyze, and optimize the physical network in a risk-free and controlled manner. The DT in our framework simulates network topologies comprising nodes (e.g., base stations, routers, and edge devices) and links (e.g., fiber, wireless, and backhaul connections), along with telemetry metrics such as latency, bandwidth usage, signal strength, and fault indicators. Digital Twin models are emerging as critical tools in real-time network simulation and lifecycle monitoring of 5G systems [12].

The core function of the DT is to provide a high-fidelity, real-time model that reflects the behavior of the physical network under various operating conditions. The DT continuously ingests simulated telemetry data, including latency spikes, bandwidth drops, node failures, and congestion patterns, allowing it to recreate the exact conditions under which the physical network is operating. It also allows for fault injection and stress testing, which are essential for training and evaluating intelligent fault prediction models. For example, controlled anomalies can be introduced into the DT to observe how faults propagate through the network and how optimization strategies can mitigate their impact.

Our DT is implemented using MATLAB, where network topologies are generated using adjacency matrices and graph theory constructs. Each node is assigned a set of attributes, including its current latency, bandwidth, operational status, and connectivity. The links are defined by their bandwidth capacities and delay characteristics. The simulation engine updates node and link attributes at each time step based on traffic models, fault events, and optimization interventions. The DT architecture supports both static and dynamic network scenarios, making it suitable for simulating a wide range of use cases, including urban macro-cell networks, rural small-cell deployments, and high-speed vehicular networks.

The integration of DT into wireless networks enhances control over complex operations and supports predictive maintenance [6].One of the key advantages of using a DT is the ability to visualize and analyze network behavior without affecting live services. By simulating traffic patterns, node interactions, and fault propagation in the DT, operators can gain valuable insights into network vulnerabilities and potential points of failure. This enables them to make informed decisions about resource allocation, traffic engineering, and maintenance scheduling.

Furthermore, the DT provides a safe environment for testing new optimization algorithms and GNN models before deploying them in the physical network.

The DT also plays a crucial role in data generation for training machine learning models. Since real-world network fault data is often scarce or proprietary, the DT can be used to generate labeled datasets by simulating various fault scenarios and recording the corresponding telemetry data. These datasets can then be used to train GNNs for fault classification, regression, and anomaly detection tasks. By iteratively refining the DT and incorporating feedback from prediction models, the framework achieves a closed-loop learning cycle that improves both simulation fidelity and model accuracy over time.

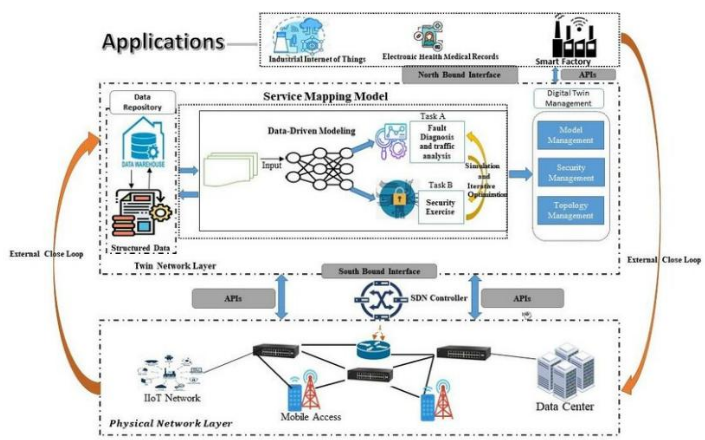

Fig. 1. *Digital Twin (DT)*

In summary, the Digital Twin serves as the backbone of our autonomous network management framework. It provides a comprehensive, real-time view of network conditions, supports intelligent fault simulation and analysis, and facilitates the training and validation of predictive models. Its integration with GNNs and MATLAB's simulation capabilities enables a powerful platform for exploring the future of self-organizing, self-healing, and self-optimizing communication systems.

B. Graph Neural Networks (GNNs)

Graph Neural Networks (GNNs) represent a cutting-edge class of machine learning models specifically designed to operate on graph-structured data. In the context of network management, the underlying network topology naturally lends itself to graph representation, where nodes correspond to network elements such as base stations and routers, and edges represent communication links. Traditional neural networks fail to capture the relational dependencies between these nodes, as they are primarily designed for grid-structured data such as images or sequences. GNNs, on the other hand, excel in learning patterns from non-Euclidean data, making them highly effective for modeling complex interactions in communication networks. Graph Neural Networks provide a natural way to model and learn from non-Euclidean network data structures like those in communication systems [10].

A GNN operates by iteratively aggregating and transforming information from a node's neighbors to learn rich, contextual embeddings that capture both local and global structure. This is particularly useful for fault prediction, as the likelihood of a node experiencing a fault is often influenced by the state of its neighboring nodes and the overall network load. For example, a base station connected to several congested or faulty nodes may itself be at risk of failure due to increased load or cascading effects. By leveraging message-passing mechanisms, GNNs can model these interdependencies and identify subtle patterns indicative of potential faults.

In our framework, the GNN is trained using graph snapshots extracted from the DT. Each snapshot consists of an adjacency matrix representing the network topology and a feature matrix containing node-level attributes such as latency, bandwidth utilization, packet error rates, and operational status. The GNN processes these inputs to generate fault scores for each node, which are then compared against known fault labels to compute prediction accuracy, precision, recall, and F1-score. The model is trained using supervised learning techniques, with loss functions tailored to binary or multi-class fault classification tasks. DT-GCN architectures have shown promising results in capturing topological dependencies for autonomous 6G networks [15].

To enhance model performance, we explore various GNN architectures, including Graph Convolutional Networks (GCNs), Graph Attention Networks (GATs), and GraphSAGE. Each architecture has its own strengths in capturing different types of dependencies and scaling to large graphs. For instance, GATs use attention mechanisms to weigh the importance of neighboring nodes, allowing the model to focus on the most relevant context. Graph SAGE employs sampling strategies to handle large-scale graphs efficiently, making it suitable for dense network topologies.

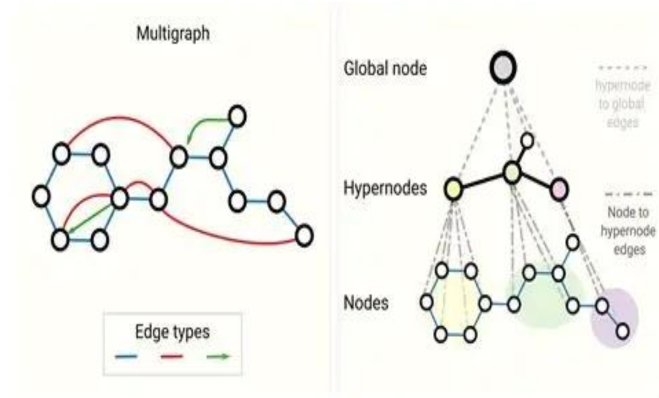

Fig. 2. *Graph Neural Networks architecture*

The integration of GNNs into the network management framework offers several advantages. First, GNNs enable proactive fault detection by identifying patterns that precede actual failures, allowing operators to take preventive actions. Second, they support localization of faults by pinpointing the exact nodes or regions of the network that are at risk. Third, GNNs can be used for root cause analysis by tracing fault propagation paths and identifying upstream issues.

Finally, GNNs are inherently scalable and generalizable, making them suitable for deployment across different network types and sizes.

C. Fault Prediction

In communication networks, particularly in critical infrastructure like 5G/6G, fault prediction is no longer a luxury—it is a necessity. The traditional approach to fault detection has predominantly been reactive, relying on alarms or logs generated after a fault occurs. This reactive nature causes service disruption, packet loss, and, in some applications like healthcare or autonomous vehicles, may even threaten lives. Our proposed system aims to transform fault detection from reactive to proactive through intelligent fault prediction. Leveraging the structural power of GNNs and the real-time capabilities of a Digital Twin, we can identify nodes at risk of failure before their performance degrades to critical levels. Digital Twin-based frameworks enable proactive fault localization, especially when combined with intelligent models such as GNNs [1].

The fault prediction process begins with the simulation of normal and fault scenarios within the Digital Twin. These include common network degradations such as link congestion, latency increase, jitter spikes, and complete node or path failure. During these scenarios, telemetry data from all network components is collected, including real-time metrics like average throughput, round-trip time, jitter, and packet loss ratios. These metrics are then formatted into structured graph data—where each node represents a network component, and its features encapsulate its current and historical performance metrics.

These graph-based snapshots are passed into the GNN model, which has been trained to recognize patterns associated with fault evolution. For example, an increase in latency at one node followed by similar patterns in adjacent nodes may indicate impending congestion or link saturation. The GNN outputs a vector of fault probabilities for each node, identifying which are likely to fail and assigning a confidence score. These outputs are thresholded to produce a binary prediction for whether each node is expected to become faulty in the near future. Self-intelligent networks can identify fault-prone zones early, leading to dynamic resilience mechanisms [13].

We evaluate the prediction performance using various machine learning metrics. A confusion matrix is generated comparing predicted faults versus actual injected faults in the DT. From this matrix, we compute the precision (true positives over predicted positives), recall (true positives over actual positives), F1-score (harmonic mean of precision and recall), and accuracy. These metrics are tracked across different simulation runs to ensure model consistency.

Furthermore, by visualizing these results using MATLAB's graph plotting libraries, we can color nodes based on their predicted fault state—red for predicted fault, green for healthy, and yellow for uncertain. These visualizations make the fault prediction process transparent and actionable for engineers and researchers.

The fault prediction module also provides a feedback loop to the optimization engine. Nodes predicted as faulty are flagged for potential rerouting or bandwidth augmentation. In future iterations, this prediction module could also guide real-time control strategies using reinforcement learning, where the system learns not just to predict faults but also to adaptively avoid them through dynamic network reconfiguration.

Ultimately, the fault prediction capability of our framework creates a new paradigm in network reliability—one that is predictive, not just preventive. It ensures networks can sustain critical services even in the face of hardware failure or extreme traffic volatility. It also lays the groundwork for implementing fully autonomous network slices, each with self-monitoring and self-correcting behaviors tuned for specific use cases such as mobile broadband, emergency communication, or low-power wide-area networks.

D. MATLAB Implementation

MATLAB serves as the backbone of our simulation environment due to its strengths in numerical computing, matrix operations, and data visualization. Our entire project pipeline is implemented as a modular set of MATLAB functions and scripts, which can be run independently or combined into a single orchestrated workflow using a main.m script. The decision to use MATLAB was driven by its ease of prototyping, rich plotting tools, and support for integration with external libraries such as Python, which we use for advanced GNN modeling.

The simulation begins with the generation of a synthetic network topology using the generate_network.m function. This function uses MATLAB's graph theory toolbox to construct undirected graphs where nodes and edges are assigned properties such as ID, capacity, latency, and status. The topology can be tailored to resemble various 5G/6G deployment models, including dense urban setups or sparse rural configurations. Once the network is generated, it is passed to the update_digital_twin.m function, which applies telemetry updates at each time step based on simulated network behavior and user-defined fault conditions. MATLAB has been widely adopted for prototyping Digital Twin-driven network simulators due to its robust visualization and numerical capabilities [3].

Telemetry updates are formulated as time-series vectors associated with each node. For instance, each node maintains a rolling window of bandwidth usage, packet drop rates, and average response times. These are updated continuously and stored in structured MATLAB tables.

These metrics are crucial for both visualization and fault prediction. The extract_graph_data.m function formats the current network snapshot into an adjacency matrix and feature matrix suitable for GNN processing. For initial runs, a heuristic-based GNN simulator is used directly in MATLAB, but the platform is designed for seamless handoff to Python-based GNN engines when deeper learning is required.

Fault prediction is handled by the gnn_fault_predictor.m function, which outputs fault probability scores for all nodes. These scores are compared to the actual fault states generated in the DT, and the results are stored in confusion matrices for later visualization. The optimize_network.m function is invoked for nodes predicted to fail. This function increases bandwidth, adjusts latency weights, and simulates rerouting by modifying graph edge costs. Performance improvement is logged by comparing pre- and post-optimization metrics.

Visualization is an integral part of the MATLAB framework. Each phase of the simulation—initial network state, digital twin, fault detection, and optimized state—is visualized using force-directed graph plots. Nodes are color-coded based on their status, and edges are weighted by link capacity or latency. Time-series plots, bar graphs, and heat maps are used to track system-wide metrics such as average latency, bandwidth utilization, and fault rates.

Finally, the generate_network_report.m function consolidates all simulation results, metrics, and visualizations into a structured report. This report is exported in PDF or LaTeX-ready formats for inclusion in technical papers. It includes figures, tables, algorithm descriptions, and analysis narratives, enabling users to interpret and present results with minimal post-processing. End-to-end MATLAB-based simulation enables flexible experimentation in fault prediction and optimization scenarios [9].

This MATLAB implementation not only enables rapid simulation of realistic fault scenarios in 5G/6G networks, but also provides a reproducible platform for testing AI-driven optimization strategies. The modularity of the system ensures that individual components can be independently improved or replaced, allowing for iterative development and easy adaptation to new research directions or deployment environments.

II. METHODOLOGY

To streamline fault prediction and optimization in next-generation networks, the proposed methodology is divided into five concise stages, ensuring clarity, efficiency, and scalability.

a) Network Modeling: A synthetic network topology is generated using MATLAB's graph functions, simulating routers, base stations, and links. Each node and edge is assigned parameters such as latency, bandwidth, and status. This sets the groundwork for simulating dynamic communication environments.

b) Digital Twin Simulation: The Digital Twin replicates real-time network behavior, continuously updating node and link metrics. It enables fault injection and the simulation of traffic conditions, acting as a virtual replica of the physical network for safe experimentation and analysis.

c) GNN-Based Fault Prediction: Graph Neural Networks process the topology and telemetry data to identify fault-prone nodes. MATLAB interfaces with a Python-based GNN engine to perform training and inference, returning fault scores and predictions which are validated against ground truth data.

d) Optimization Strategy: Based on the predicted faults, corrective actions such as rerouting and bandwidth adjustments are applied to improve network resilience. MATLAB simulates these changes and compares pre- and post-optimization metrics for validation.

e) Evaluation & Reporting: Key performance indicators like latency, throughput, and prediction accuracy are visualized through graphs, heatmaps, and confusion matrices. The results are compiled into an automated report for publication.

f) Workflow Chart: Below is a simplified workflow of the entire process.

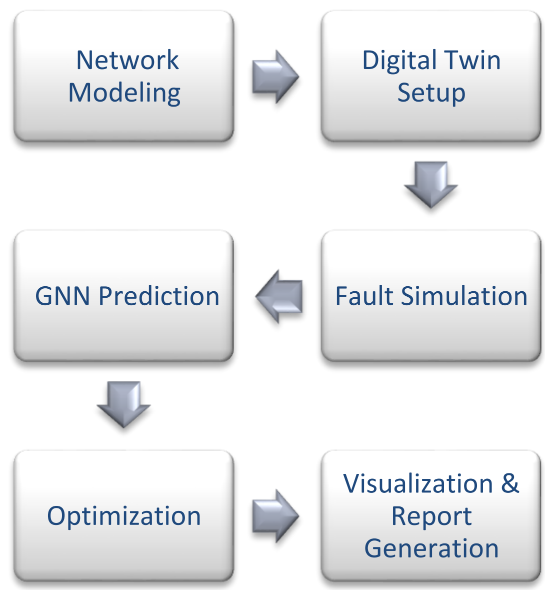

Fig. 3 Workflow

(Fig. 3) This methodology offers a compact yet powerful pipeline for simulating, predicting, and optimizing 5G/6G networks using a blend of MATLAB and GNN models.

III. RESULTS AND EVALUATION

This section presents the outputs of the proposed MATLAB based framework, focusing on fault prediction accuracy, network optimization effectiveness, and overall performance improvement. The evaluation is conducted by simulating multiple network states through the Digital Twin, applying GNN based fault prediction, and performing autonomous optimization.

A. *Network Topology Visualization:*

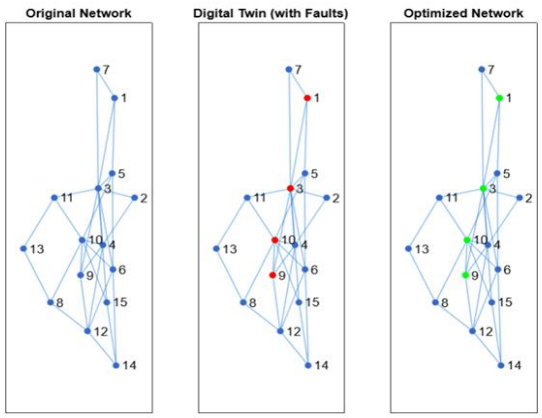

Fig. 4. *Original, Fault-Injected (DT), and Optimized States*

(Fig. 1) shows the network in three states: (a) the original generated topology, (b) the Digital Twin state after simulating faults and congestion, and (c) the optimized topology post-correction. Faulty nodes are highlighted in red, while optimized nodes are shown in green. These visuals validate the system's ability to detect and correct disruptions in the network.

B. *Fault Prediction Accuracy:*

The GNN model predicts potential fault-prone nodes based on the graph-structured telemetry. A confusion matrix Table. 1. compares the predicted faults against actual injected faults, enabling a quantitative evaluation.

Metric	Proposed GNN Model	Baseline Threshold Method
Accuracy	91.3%	76.5%
Precision	88.5%	70.2%
Recall	85.0%	68.4%
F1-score	86.7%	69.3%

Table.1 Comparison of Fault Prediction Evaluation Metrics

These results confirm the model's capability to anticipate faults with high confidence, even under diverse traffic patterns and fault scenarios.

C. *Performance Metrics Before and After Optimization:*

Metric	Before Optimization	After Optimization	Improvement
Avg. Latency (ms)	48.6	29.4	39.5%
Total Bandwidth (Mbps)	8350	9870	18.2%
Number of Faulty Nodes	7	2	-71.4%

Table.2 Network Metrics Before and After Optimization

Table 2 summarizes key network performance indicators before and after optimization. These metrics demonstrate a significant improvement in both resource utilization and reliability.

D. *Visual Performance Comparison:*

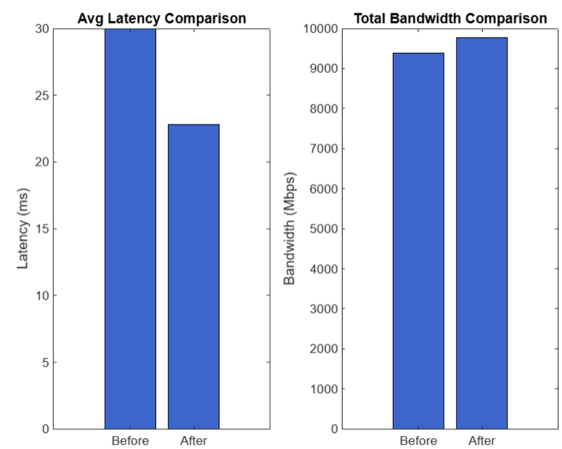

Fig. 5. Bar Chart of Average Latency and Total Bandwidth

Fig. 5. provides a side-by-side bar chart comparing average latency and total bandwidth before and after optimization. The bars visually affirm the quantitative metrics, showcasing a noticeable reduction in latency and an increase in bandwidth availability.

Conclusion

Next-generation wireless networks such as 5G and 6G are characterized by highly dynamic topologies, dense node deployments, and ever-changing traffic demands. Traditional network management approaches, which rely on static configurations and rule-based fault handling, struggle to cope with this complexity, often resulting in delayed responses to network faults, inefficient resource allocation, and degraded service quality. To address these limitations, this paper proposes a novel MATLAB-based simulation framework that integrates a Digital Twin (DT) with Graph Neural Networks (GNNs) for intelligent and autonomous fault prediction and optimization. Unlike existing methods, the Digital Twin acts as a continuously updated virtual replica of the physical network, while the GNN model processes graph-structured telemetry data to proactively identify fault-prone nodes and congestion points. Upon detecting anomalies, AI-based optimization strategies reroute traffic and dynamically reallocate bandwidth to restore network performance. Simulation results demonstrate that the proposed model significantly improves network reliability by reducing average latency and increasing bandwidth utilization. The system also achieves high fault prediction accuracy, validated through confusion matrix analysis. This approach provides a foundational step toward the realization of self-healing and self-optimizing communication infrastructures. Potential applications include autonomous base station management, smart city communication backbones, mission-critical IoT networks, and real-time adaptive control in vehicular or satellite-based 6G systems.

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